

CURRICULUM VITAE

GUSTAVO MASTROROCCHO MARQUES

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Educational Information

- 2002-2006 B. S. in Oceanography, University of Sao Paulo, Sao Paulo, Brazil.
2007-2010 M. S. in Physical Oceanography, University of Massachusetts Dartmouth,
Dartmouth, MA, USA. Advisor: Prof. Wendell Brown.
Thesis title: *Secondary Flow Associated with Transient Tidal Eddy Motion in the
Western Gulf of Maine.*
2010-2015 Ph. D. in Meteorology & Physical Oceanography, University of Miami,
Miami, FL, USA. Advisor: Prof. Tamay Özgökmen.
Dissertation title: *On the processes controlling Antarctic dense shelf water outflows.*

Work History

- 2003-2006 Research assistant, Oceanographic Institute, University of Sao Paulo, Sao Paulo,
Brazil.
2007-2010 Research assistant, School for Marine Science and Technology, University of
Massachusetts Dartmouth, Dartmouth, MA.
2010-2015 Research and teaching assistant, Meteorology & Physical Oceanography,
University of Miami, Miami, FL.
2015-2017 Postdoctoral Research Associate, Program in Atmospheric and Oceanic
Sciences, Princeton University / Geophysical Fluid Dynamics Laboratory,
Princeton, NJ.
2017-2022 Project Scientist I, Oceanography Section, Climate and Global Dynamics
Division, National Center for Atmospheric Research, Boulder, CO.
2022-2024 Project Scientist II, Oceanography Section, Climate and Global Dynamics
Division, National Center for Atmospheric Research, Boulder, CO.
2024-2025 Scientist II, Oceanography Section, Climate and Global Dynamics
Division, National Center for Atmospheric Research, Boulder, CO.
2024-2025 Scientist II, Oceanography Section, Climate and Global Dynamics
Division, National Center for Atmospheric Research, Boulder, CO.
2017-present Visiting Scientist, Geophysical Fluid Dynamics Laboratory, Princeton, NJ.
2025-present Scientist V, Oceanography Section, Climate and Global Dynamics
Division, National Center for Atmospheric Research, Boulder, CO.

Professional Schools Attended

1. *Data Processing*, Escola Tecnica Oswaldo Cruz, Sao Paulo, Brazil, 1995-1998.
2. *Modeling the Ocean in the Climate System*, III Brazilian Symposium of Oceanography. Sao Paulo, Brazil, December 04-08, 2006.
3. *Physical and Biological Ocean Modeling*, Universidad de Concepcion winter school on Physical and Biological Ocean Modeling, Dichato, Chile, July 03-14, 2007.
4. *Glider Training Course*, Coastal Ocean Observation Laboratory, Rutgers University, New Brunswick, NJ, January 9-11, 2008.
5. *Coastal Modeling Summer School*, University of Toulon/CNRS, La Londe des Maures, France, September 18-23, 2011.
6. *ROMS/TOMS User Workshop*, Rio de Janeiro, Brazil, October 22-25, 2012.
7. *Fluid Dynamics of Sustainability and the Environment Summer School*, Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Cambridge, UK, September 1-12, 2014.
8. *Using Satellite Observations to Advance Climate Models*, Keck Institute for Space Studies, California Institute of Technology, Pasadena, CA, August 31-September 4, 2015.
9. *Artificial Intelligence for Earth System Science Summer School*, National Center for Atmospheric Research, Boulder, CO, June 22-26, 2020.
10. *Google Earth Engine Workshop*. Boulder, CO, October 28th, 2022.
11. *Tackling Climate-Induced Challenges with AI* (fellow), Virtual six-week boot camp organized by the Convergence Research (CORE) Institute, March 6th to April 15th, 2023.
12. *Machine Learning for Environmental Engineering*, School of Engineering and Applied Science, Columbia University, Auditor, Fall 2023.

Honors and Awards

Scientific initiation scholarship from the Foundation for Research Support of the State of Sao Paulo (FAPESP), 2006.

Convergence Research (CORE) Institute Fellow, University of California, San Diego (UC San Diego), July 10–14, 2023.

Professional Memberships

American Geophysical Union.

Research Grants Awarded

1. National Science Foundation, Division of Ocean Sciences, Collaborative Research: Ocean Transport and Eddy Energy. September 2019 – August 2027, \$693,164 (NCAR portion), Principal Investigator.
2. National Science Foundation, Directorate for Computer and Information Science and Engineering, Office of Advanced Cyberinfrastructure, Collaborative Research: Frameworks: A community platform for accelerating observationally-constrained regional oceanographic modeling. September 2023 – August 2028, \$3,358,308 (NCAR portion), Co-Principal Investigator.
3. NOAA Climate Program Office, Assessing and Improving the Representation of Riverine Influences on Large Marine Ecosystems in Regional MOM6 Models. May 2025 – April 2028, \$249,537 (NCAR portion), Principal Investigator.
4. National Science Foundation, Office of Advanced Cyberinfrastructure, Collaborative Research: Frameworks: TURBO: Towards Ultra-high Resolution Community Earth System Model (CESM) with MOM6 and Ocean Biogeochemistry. July 2025 – June 2030, \$4,499,035 (NCAR portion), Co-Principal Investigator.

Community Service

Co-Chair of the Ocean Model Working Group (OMWG) of the Community Earth System Model (CESM) and lead developer of the Modular Ocean Model version 6 (MOM6) within CESM. Contributed to the development of new ocean parameterizations, model improvements, and diagnostic tools widely adopted by the university and research communities. Engage extensively with national and international collaborators, providing guidance on ocean modeling and experimental design. Lecturer for the annual CESM Tutorial since 2019, teaching the fundamentals of ocean modeling.

Committee and Panel Service

Member, *Task Team for Ice Sheet Model Intercomparison Project for CMIP6 Ocean Forcing*, January-December 2018.

Member, *the Community Earth System Model Tutorial*, November 2018 – present.

Chair, *the Community Earth System Model Tutorial*, September 2020 – October 2021.

Member, CGD/NCAR - *Community Earth System Model Dycore Evaluation Panel*, October 2021 – February 2022.

Member, *Steering Committee for Ocean Transport and Eddy Energy Climate Process Team*, September 2023 – August 2025.

Member, Coupled Prediction Task Team - an international effort within the OceanPredict science program to coordinate and improve global and regional ocean analysis and forecasting systems, August 2023 – present.

Chair, Organizing Committee for the joint CLIVAR/COMMODORE Workshop: Ocean Model Development, Data-driven Parameterizations, and Machine Learning in Ocean Models of the Earth System. National Center for Atmospheric Research, Boulder, CO, September 09-12, 2024.

Member, *the CLIVAR Working Group on Ocean Model Development Panel (OMDP)*, September 2025 – present.

Editorial Board Member, Special Issue “*In memory of William Kurt Dewar: Exploring the dynamics of oceanic boundary currents (e.g., the Gulf Stream) and their impact on weather,*” *Frontiers in Marine Science*, 2025–present.

Member, Organizing Committee for the Regional Ocean Modeling with MOM6 in the Community Earth System Model Framework Workshop. National Center for Atmospheric Research, Boulder, CO, October 13-17, 2025.

Reviewer, NCAR Advanced Study Program Postdoctoral Applications (ongoing).

Professional Meetings

Organizer, Mesoscale Eddy Energy and Ocean Transport, *Ocean Sciences Meeting*, 24 February – 04 March, Online, 2022.

Session Moderator, Multiscale Eddy Dynamics and Tracer Transport: Bridging Observations, Theory, and Modeling, *Ocean Sciences Meeting*, 18-23 February, New Orleans, LA, 2024.

Professional Reviews

For Journals: Journal of Physical Oceanography, Ocean Modelling, Ocean Science, Journal of Geophysical Research - Oceans, Journal of Advances in Modeling Earth Systems, Earth and Planetary Science Letters, Deep-Sea Research Part I, Geophysical Research Letters, Geoscientific Model Development.

For Agencies: National Science Foundation Antarctic Integrated System Science (AISS) program, the Natural Environment Research Council.

Educational and Outreach Activities

Lecturer, *Geophysical Fluid Dynamics I*, RSMAS/MPO-511. University of Miami, Fall 2011, 2012 and 2013.

Lecturer, *Atmospheric and Oceanic Turbulence*, RSMAS/MPO-664. University of Miami, Fall 2013.

Lecturer and organizer, *Introduction to the Regional Ocean Modeling System (ROMS)*. University of Miami, Fall 2013.

Lecturer, *Ocean Physics for Climate*, GEO-425, Princeton University, Fall 2015.

Organizer, *Gender Equity in Geosciences*, seminar by Dr. Sonya Legg, Boulder, CO, October 04, 2023.

Dissertation Committee, Giovanni Seijo-Ellis, Atmospheric and Oceanic Sciences, University of Colorado Boulder, 2020-2024.

Member, Organizing Committee, *Regional ocean modeling with MOM6 in the Community Earth System Model framework*, Boulder, CO, October 13-17, 2025.

Lecturer, *Ocean Modeling I* at the CESM Tutorial, National Center for Atmospheric Research, since Summer 2019 to present.

Mentorship and Supervision

Theresa Morison, NCAR Graduate Student Advanced Study Program, *co-host*, 2018-2019.

Xiaoning Wu, NCAR Graduate Student Advanced Study Program, *co-host*, 2019-2021.

Kenzie Klisz, NSF NCAR Earth System Science Internship, *mentor*, Summer 2022.

Maurício Rebouças Rocha, Graduate Student Fullbright Program, *host*, 2022-2023.

Giovanni Seijo-Ellis, NCAR Graduate Student Advanced Study Program, *host*, 2021-2024.

Xinru Li, Postdoc at Princeton University/NOAA-GFDL, *co-mentor (informal)*, 2022-2024.

Houssam Yassin, Postdoc at Colorado University Boulder, *co-mentor (informal)*, 2023-2025.

Benjamin Barr, Scientist III, *supervisor*, 2026-present.

Chengzhu (William) Wu, Scientist III, *supervisor*, 2026-present.

Invited Presentations

1. Marques, G. M. and W.S. Brown, 2010: Transient Tidal Eddy Motion and Associated Secondary Flow in the Western Gulf of Maine. *Graduate School of Oceanography*, University of Rhode Island, Rhode Island, USA, June 10.
2. Marques, G. M., M. Harrison, O. Sergienko, and R. Hallberg, 2016: Changes in bottom water formation in the western Ross Sea due to the melting of ice shelves in West Antarctica. *Rising Coastal Seas on a Warming Earth II*, Abu Dhabi, United Arab Emirates, May 16-18.
3. Marques, G. M., 2017: Processes controlling bottom water formation and sub-ice-shelf melting in idealized simulations of Antarctic margins. *Center for Coastal Physical Oceanography*, Old Dominion University, Norfolk, VA, USA, March 24.

4. Marques, G. M., A. Stern, M. Harrison, O. Sergienko, A. Adcroft, and R. Hallberg, 2018: The impact of wind forcing on sub-ice-shelf melting and circulation. *Rising Coastal Seas on a Warming Earth III*, Abu Dhabi, United Arab Emirates, May 7-9.
5. Marques, G. M., 2018: New developments in ocean models since CMIP5. *ISMIP6 Workshop*, Sassenheim, Netherlands, September 10-14.
6. Marques, G. M., 2021: Towards CESM3 with MOM6. *Earth System Model Working Group*, Institute for Marine and Atmospheric Research, Utrecht University, Virtual, February 18.
7. Marques, G. M., 2021: Conceitos básicos de modelagem oceânica com foco nos modelos do CESM (in portuguese). IV Semana Meteorológica da UFSM. Centro de Ciências Naturais e Exatas, Universidade Federal de Santa Maria, Virtual, June 10.
8. Marques, G. M., 2021: Improving the representation of oceanic processes in NCAR's next-generation climate model. The Department of Atmospheric and Oceanic Sciences Colloquium. University of Colorado, Boulder, CO, October 29.
9. Marques, G. M., 2022: An overview of the CESM/MOM6 development. Presented during the Model Diagnostics Task Force monthly telecon. Virtual, October 3.
10. Marques, G. M., 2023: Assessing the performance of a machine-learning-based parameterization of eddy kinetic energy in global ocean/sea-ice simulations. Keynote seminar presented at the 2023 DRAKKAR Ocean Modelling Workshop. Grenoble, France, January 30.
11. Marques, G. M., 2023: The state of CESM with focus on the ocean component and a touch of machine learning. M2LInES. Multiscale Machine Learning In Coupled Earth System Modeling Spring meeting. Edith Macy Center, Briarcliff Manor, NY, June 5.
12. Marques, G. M., 2024: MOM6 Developments Towards CESM3: Advancements, Challenges, and Future Directions. NOAA's Subseasonal and Seasonal Applications Workshop. Online. September 6.
13. Marques, G. M., 2025: Overview of Energy Backscatter in CESM3. Eddy Energy Climate Process Team Meeting. Courant Institute of Mathematical Sciences, New York, NY, August 13.

Publications

A. Thesis

1. Marques, G. M., 2010: Secondary flow associated with transient tidal eddy motion in the western Gulf of Maine. Master Thesis, Department of Estuarine and Ocean Science, University of Massachusetts Dartmouth, Dartmouth, MA, 116 pp.

B. Dissertation

1. Marques, G. M., 2015: On the Processes Controlling Antarctic Dense Shelf Water Outflows. PhD. Dissertation, Department of Meteorology & Physical Oceanography, Rosenstiel School of Marine & Atmospheric Science, University of Miami, Miami, FL, 137 pp.

C. Refereed Journal Articles (* From Dissertation or Thesis Work)

1. *Brown, W. S., and G. M. Marques, 2013: Tidal eddy motions in the western Gulf of Maine, Part 1: Primary Structure. *Contin. Shelf Res.*, **63**, S90-S113, DOI: 10.1016/j.csr.2012.08.018.
2. *Marques, G. M., and W. S. Brown, 2013: Tidal eddy motions in the western Gulf of Maine, Part 2: Secondary Flow. *Contin. Shelf Res.*, **63**, S114–S125, DOI: 10.1016/j.csr.2012.02.008.
3. *Marques, G. M., L. Padman, S. R. Springer, S. L. Howard, and T. Özgökmen, 2014: Topographic vorticity waves forced by Antarctic dense shelf water outflows. *Geophys. Res. Lett.*, **41**, 1247–1254, DOI: 10.1002/2013GL059153.
4. *Marques, G. M., and T. Özgökmen, 2014: On modeling turbulent exchange in buoyancy-driven fronts. *Ocean Modelling*, **83**, 43-62, DOI: 10.1016/j.ocemod.2014.08.006.
5. *Marques, G. M., M. G. Wells, L. Padman, and T. Özgökmen, 2017: Flow splitting in numerical simulations of oceanic dense-water outflows. *Ocean Modelling*, **113**, 66-84, DOI: 10.1016/j.ocemod.2017.03.011.
6. Stern, A., A. Adcroft, O. Sergienko, and G. M. Marques, 2017: Modeling tabular icebergs submerged in the ocean. *J. Adv. Model. Earth Sys.*, **9**, 1948–1972, DOI: 10.1002/2017MS001002.
7. Wu, X., K. A. Reed, C. L. P. Wolfe, G. M. Marques, S. D. Bachman, F. O. Bryan, 2021: Coupled aqua and ridge planets in the community earth system model. *J. Adv. Model. Earth Sys.*, **13**, e2020MS002418, DOI: 10.1029/2020MS002418.
8. Wu, X., K. A. Reed, C. L. P. Wolfe, G. M. Marques, S. D. Bachman, F. O. Bryan, 2021: The Dependence of Tropical Modes of Variability on Zonal Asymmetry. *Geophysical Research Letters*, **48**, p.e2021GL093966, DOI: 10.1029/2021GL093966.
9. Grooms, I., N. Loose, R. Abernathey, J. M. Steinberg, S. D. Bachman, G. M. Marques, A. P. Guillaumin, E. Yankovsky, 2021: Diffusion-based smoothers for spatial filtering of gridded geophysical data. *J. Adv. Model. Earth Sys.*, **13**, e2021MS002552, DOI: 10.1029/2021MS002552.
10. Loose, N., R. Abernathey, J. Busecke, J. M. Steinberg, I. Grooms, E. Yankovsky, E. G. M. Marques, H. Khatri, S. D. Bachman, 2021: GCM-Filters: A Python package for Diffusion-based Spatial Filtering of Gridded Data from General Circulation Models. *The Journal of Open Source Software*, **7(70)**, 3947, DOI: 10.21105/joss.03947.
11. Partee, S., M. Ellis, A. Rigazzi, A. Shao, S. D. Bachman, G. M. Marques, B. Robbins, 2022: Using Machine Learning at scale in numerical simulations with SmartSim: An application to

ocean climate modeling. *Journal of Computational Science*, **62**, 101707, DOI: 10.1016/j.jocs.2022.101707.

12. Marques, G. M., N. Loose, E. Yankovsky, J. M. Steinberg, C. Y. Chang, N. Bhamidipati, A. Adcroft, B. Fox-Kemper, S. M. Griffies, R. W. Hallberg, M. F. Jansen, 2022: NeverWorld2: An idealized model hierarchy to investigate ocean mesoscale eddies across resolutions. *Geoscientific Model Development*, **15**, 6567–6579, DOI: 10.5194/gmd-15-6567-2022.
13. Marques, G. M., A. Shao, S. D. Bachman, G. Danabasoglu, F. Bryan, 2023: Representing Eddy Diffusion in the Surface Boundary Layer of Ocean Models with General Vertical Coordinates. *J. Adv. Model. Earth Sys.*, **15**(6), e2023MS003751, DOI: 10.1029/2023MS003751
14. Loose, N., G. M. Marques, A. Adcroft, S. D. Bachman, S. M. Griffies, I. Grooms, R. W. Hallberg, M. F. Jansen, 2023: Comparing two parameterizations for the restratification effect of mesoscale eddies in an isopycnal ocean model. *J. Adv. Model. Earth Sys.*, **15**(12), e2022MS003518, DOI: 10.1029/2022MS003518.
15. Seijo-Ellis, G., D. Giglio, G. M. Marques, and F. Bryan, 2024: CARIB12: A regional Community Earth System Model/Modular Ocean Model 6 configuration of the Caribbean Sea. *Geoscientific Model Development*, **17**, 8989–9021, DOI:10.5194/gmd-17-8989-2024.
16. Grooms, I., N. Agarwal, G. M. Marques, P. J. Pegion, and H. Yassin, 2025: The stochastic GM + E closure: A framework for coupling stochastic backscatter with the Gent and McWilliams parameterization. *J. Adv. Model. Earth Sys.*, **17**(5), e2024MS004560, DOI:10.1029/2024MS004560.
17. Uchida, T., A. Bodner, B. G. Reichl, A. J. Adcroft, B. Fox-Kemper, M. Ilicak, M. Bentsen, G. M. Marques, and W. G. Large, 2026: Representation of surface mixed-layer eddies affects the large-scale ventilation of the global ocean. *Geophysical Research Letters*, **53**(4), e2025GL116872, DOI:10.1029/2025GL116872.
18. Yassin, H., G. Marques, and I. Grooms, 2026: Balancing backscatter and diffusion in a 1/4° forced global ocean model. *J. Adv. Model. Earth Sys.*, **18**(3), e2025MS005462 doi:10.1029/2025MS005462.
19. Rocha, M. R., G. M. Marques, F. S. Castruccio, A.-L. Deppenmeier, O. T. Sato, and G. Danabasoglu, 2026: Mechanisms controlling the past, present, and future trends in the South Atlantic Ocean's heat storage in a set of CESM2 large ensemble simulations. *Journal of Climate*, DOI:10.1175/JCLI-D-25-0089.1.
20. Yung, C. K., X. S. Asay-Davis, A. Adcroft, C. Y. S. Bull, J. De Rydt, M. S. Dinniman, B. K. Galton-Fenzi, D. Goldberg, D. E. Gwyther, R. Hallberg, M. Harrison, T. Hattermann, D. M. Holland, D. Holland, P. R. Holland, J. R. Jordan, N. C. Jourdain, K. Kusahara, G. M. Marques, P. Mathiot, D. Menemenlis, A. K. Morrison, Y. Nakayama, O. Sergienko, R. S.

Smith, A. Stern, R. Timmermann, and Q. Zhou, 2026: Results of the second Ice Shelf–Ocean Model Intercomparison Project (ISOMIP+), *The Cryosphere*, **20**, 2053–2088, DOI: 10.5194/tc-20-2053-2026.

21. Morrison, T. J., F. Bryan, J. L. McClean, S. T. Gille, and G. M. Marques, 2026: A glacial fjord box model: Derivation and representation of glacially modified water. *J. Adv. Model. Earth Sys.*, DOI: 10.1029/2024MS004794.

D. Other External Refereed Publications

1. Pearse, S., and Marques, G. M. Marques, 2023: Visualizing NeverWorld2: A set of ocean simulations to study mesoscale eddies across resolutions. Proceedings of the 2023 ACM SIGGRAPH Symposium on Visualization, ACM, DOI: 10.1145/3569951.3603640.
2. Marques, G. M., A. Adcroft, G. Danabasoglu, F. Lemarié, and J. McClean, 2025: Ocean model development, data-driven parameterizations, and machine learning in ocean models of the Earth system: A U.S. CLIVAR report. *U.S. CLIVAR*, **2025-3**, 31 pp., DOI:10.5065/bxk3-gs26.

E. Publications Submitted or in Preparation

1. Danabasoglu, G., F. S. Castruccio, B. Boza, A. M. Barthel, A. Biastoch, A. Blaker, A. Bozec, D. Bruciaferri, F. O. Bryan, Y. Fu, I. Grooms, C. Guiavarc'h, H. Hayashida, A. McC. Hogg, R. M. Holmes, D. Iovino, A. E. Kiss, M. S. Lozier, G. M. Marques, A. Megann, F. U. Schwarzkopf, D. Storkey, L. van Roekel, J. Wolfe, X. Xu, and R. Zhang, 2025: Simulated and observed transport estimates across the Overturning in the Subpolar North Atlantic Program (OSNAP) section. doi:10.5194/egusphere-2025-5406. (submitted)

F. Non-refereed Publications and Presentations

1. Marques, G. M., and J. Harari, 2006: Assessment of the near field associated with the Santos sewage outfall using a non-hydrostatic model and fecal coliform analysis. *III Brazilian Symposium of Oceanography*, Sao Paulo, Brazil, December 4-8 (poster).
2. Marques, G. M., and W. S. Brown, 2009: Transient tidal eddy motion in the Western Gulf of Maine: preliminary dynamical results. *Gordon Research Conference on Coastal Ocean Circulation*, New London, NH, June 8-12 (poster).
3. Marques, G. M., and W. S. Brown, 2010: Secondary Circulation Associated with Strong Tidal Flow in the Western Gulf of Maine. *Ocean Sciences Meeting*, Portland, Oregon, USA. February 22-26 (poster).
4. Marques, G. M., and T. M. Özgökmen, 2012: Lagrangian Coherent Structures Introduced by Overflows. *5th Lagrangian Analysis and Prediction of Coastal and Ocean Dynamics (LAPCOD)*, Miami Beach, FL, USA. June 11-15 (poster).

5. Marques, G. M., and T. M. Özgökmen, 2012: On modeling the turbulent exchange in buoyancy-driven fronts. *ROMS/TOMS User Workshop*, Rio de Janeiro, Brazil. October 22-25.
6. Marques, G. M., T. M. Özgökmen, 2011: Comparing mixing and coherent turbulent features using an OGCM and a non-hydrostatic spectral element model. *Coastal Modeling Summer School*, University of Toulon in La Londe des Maures, France. September 18-23 (poster).
7. Marques, G. M., L. Padman and T. M. Özgökmen, 2013: Idealized Numerical Model Simulations of AABW Production in the NW Ross Sea: Sensitivity to Grid Resolution, Mixing Models and Background Stratification. *Gordon Research Conference on Coastal Ocean Circulation*, Biddeford, ME, June 9-14 (poster).
8. Marques, G. M., L. Padman, S. R. Springer, S. L. Howard, and T. M. Özgökmen, 2014: Topographic vorticity waves forced by Antarctic dense shelf water outflows. *Ocean Science Meeting*, Honolulu, HI, February 23-28.
9. Marques, G. M., L. Padman and T. M. Özgökmen, 2016: Flow Splitting in Numerical Simulations of Oceanic Dense-Water Outflows. *Ocean Science Meeting*, New Orleans, LA, USA. February 21-26.
10. Marques, G. M., A. Stern, M. Harrison, O. Sergienko, A. Adcroft, and R. Hallberg, 2017: Sensitivity of sub-ice-shelf melting to changes atmospheric forcing. *Regional Sea Level Changes and Coastal Impacts*, New York, NY, USA, July 10.
11. Marques, G. M., A. Stern, M. Harrison, O. Sergienko, and R. Hallberg, 2017: The effects of sub-ice-shelf melting on dense shelf water formation and export in idealized simulations of Antarctic margins. *European Geosciences Union General Assembly*, Vienna, Austria, April 23-28.
12. Marques, G. M., 2018: Coupling MOM6 into CESM: Progress, challenges, and next steps. *Ocean Model Working Group*, CESM Winter Meeting, Boulder, CO, January 11.
13. Marques, G. M., A. Stern, M. Harrison, O. Sergienko, A. Adcroft, and R. Hallberg, 2018: The impact of sub-ice-shelf melting on sea-ice formation under different atmospheric forcing. *Ocean Science Meeting*, Portland, OR, February 15.
14. Marques, G. M., 2018: Improving vertical mixing parameterizations in MOM6. *Ocean Model Working Group*, 23rd Annual CESM Workshop, Boulder, CO, June 19.
15. Marques, G. M., O. Sergienko, M. Harrison, A. Adcroft, and R. Hallberg, 2018: An idealized framework to study interactions between ocean, sea ice and ice shelves. *COMMODORE Workshop*, Paris, France, September 18.

16. Marques, G. M. and G. Leguy, 2019: Standalone and coupled MISOMIP experiments using CISM and MOM6. *Land Ice Working Group Meeting*, CESM Winter Meeting, Boulder, CO, USA, February 4.
17. Marques, G. M., 2019: Developing MOM6 towards CESM3. *Ocean Model Working Group*, 24th Annual CESM Workshop, Boulder, CO, June 8.
18. Marques, G. M., 2020: Representing the horizontal diffusion of tracers within the boundary layers of general vertical coordinate ocean models. *Hamburg COMMODORE Conference*, Hamburg, Germany, January 28-31.
19. Marques, G. M., and S. D. Bachman, G. Danabasoglu, and F. O. Bryan, 2020: The effects of representing unresolved ocean variability using an energy backscatter approach on climate-relevant oceanic metrics. *Surface Currents in the Coupled Ocean-Atmosphere System Workshop (US CLIVAR)*, San Diego, CA, February 22-23 (poster).
20. Marques, G. M. 2020: Current simulations with MOM6 in CESM. *CESM Ocean Working Group & MOM6 Webinar Series*, Virtual, April 3.
21. Marques, G. M. 2020: Representing the mesoscale horizontal diffusion of tracers within the boundary layers of general vertical coordinate ocean models. *CESM Ocean Working Group & MOM6 Webinar Series*, Virtual, June 8.
22. Marques, G. M., and S. D. Bachman, 2021: Enabling Intelligent parameterizations through distributed, online inference with SmartSim: A case study simulating ocean eddy kinetic energy in MOM6. *CGD Seminar Series*, Virtual, February 2.
23. Marques, G. M., 2021: An Update on the CESM MOM6 Effort. *Joint Session on Ocean-Ice Sheet Interaction with Land Ice Working Group*, CESM Winter Meeting, Virtual, February 3.
24. Asay-Davis, X., Bull, C. Y. S., Cornford, S., Cougnon, E., De Rydt, J., Galton-Fenzi, B. K., Gladstone, R., Goldberg, D., Gwyther, D., Jordan, J., Jourdain, N., Leguy, G., Lipscomb, W., Marques, G. M., Martin, D. F., Nakayama, Y., Naughten, K. A., Smith, R. S., Seroussi, H., and Zhao, C., 2021: Analysis of the Marine Ice Sheet-Ocean Model Intercomparison Project first phase (MISOMIP1), *European Geosciences Union General Assembly*, Virtual, April 19–30.
25. Bailey, D. A., M. M. Holland, G. M. Marques, 2021: Coupling of sea ice and ocean components in Earth System Models. *6th Conference on Polar Meteorology and Oceanography Virtual Meeting*, June 1 (poster).
26. Marques, G. M., 2021: An Update on the CESM MOM6 Effort. *Joint Session on Ocean-Ice Sheet Interaction with Land Ice Working Group*, CESM Workshop, Virtual, June 3.
27. Marques, G. M., 2022: Groundwork for incorporating enthalpy fluxes in CESM-MOM6. *CESM Ocean Model Work Group Meeting*, Boulder, CO, January 25.

28. Bachman, S., A. Shao, G. M. Marques, J. Early, and S. Keating, 2022: A dynamic vertical grid for the optimal resolution of baroclinic modes in ALE ocean models. *Ocean Sciences Meeting*, March 1 (virtual poster).
29. Adcroft, A., N. Loose, G. M. Marques, J. Steinberg, E. Yankovsky, and L. Zanna, 2022: NeverWorld2: An idealized model hierarchy and dataset for studying mesoscale eddies. *Ocean Sciences Meeting*, March 3 (virtual poster).
30. Marques, G. M., S. D. Bachman, A. Adcroft, M. Jansen, G. Danabasoglu, and L. Zanna, 2022: Sensitivity of Climate-Relevant Metrics to Thickness Diffusivity in an Idealized Model. *Ocean Sciences Meeting*, March 3 (virtual poster).
31. Marques, G. M., A. Shao, S. D. Bachman and others, 2022: Assessing the performance of a machine-learning-based parameterization of eddy kinetic energy in global ocean/sea-ice simulations. Climate Process Team. *Climate Process Team: Ocean Transport and Eddy Energy Annual Meeting*, Boulder, CO, April 13.
32. Marques, G. M., A. Shao, S. D. Bachman and others, 2022: A method for applying horizontal surface eddy diffusion in ocean models with a general vertical coordinate. *Climate Process Team: Ocean Transport and Eddy Energy Annual Meeting*, Boulder, CO, April 14.
33. Marques, G. M., 2022: Status of fully coupled simulations with MOM6. *Ocean Model Working Group, Annual CESM Workshop*, Boulder, CO, June 8.
34. Marques, G. M., 2023: Progress and Plans - CESM/MOM6. *CESM Ocean Model Work Group Meeting*, Boulder, CO, February 9.
35. Marques, G. M., E. Yankovsky, S. D. Bachman, I. Grooms, P. Gent, G. Danabasoglu, 2023: Backscatter parameterizations in global eddy-permitting simulations with CESM. *Ocean Transport and Eddy Energy Annual Meeting*, Woods hole, MA, May 9.
36. Marques, G. M., 2023: Progress towards CESM3 ocean model and workflows, *Ocean Model Working Group, Annual CESM Workshop*, Boulder, CO, June 13.
37. Marques, G. M., 2024: Exploring the power of machine learning to tune tracer diffusivities offline. *CESM Ocean Model Work Group Meeting*, Boulder, CO, February 8.
38. Marques, G. M., 2024: Towards the CESM3 ocean component. *CESM Ocean Model Work Group Meeting*, Boulder, CO, February 8.
39. Marques, G. M., 2024: Status of MOM6 towards CESM3 Release & Discussion, *Ocean Model Working Group, Annual CESM Workshop*, Boulder, CO, June 10.

40. Marques, G. M., 2024: Ocean vertical grid decision. *CESM Community Advisory Board (CAB), Scientific Steering Committee (SSC), and Working Group (WG) Cochairs Meeting*, Boulder, CO, June 13.
41. Marques, G. M., 2024: The Effects of Different Enthalpy Formulations on the Surface Heat Budget of an Earth System Model. *American Geophysical Union Fall Meeting*, Washington, D.C., December 9–13.
42. Marques, G. M. & Grooms, I., 2026: CESM 3 Ocean Overview. *CESM Ocean Model Work Group Meeting*, Boulder, CO, February 5.
43. Marques, G. M., et al., 2026: The Ocean Component of CESM3: Key Advancements, Challenges, and Coupled Model Evaluation. *Ocean Sciences Meeting*, Glasgow, Scotland, February 22–27 (poster).

Experience at Sea (summary)

Coastal cruises around Sao Paulo and Rio de Janeiro, Brazil, for observation of physical, biological, geological and chemical properties. Coastal cruises around Massachusetts, USA. Instruments deployed: glider, CTD, ADCP, rosette, Niskin and Nansen bottles, fluorometer, plankton net, Van Veen grab, box-corer, fish net and side-scan sonar.

Computational Modeling Experience

MOM6 (developer), ROMS, MITgcm, QUODDY, POP2, FLUIDITY, POM and SWAN.

Computer Skills

- Data base: MySQL.
- Languages: Python, Fortran, C++, Shell scripts (Bash and C shell), NCL, PHP, Awk and JavaScript.
- Applications: Generic Mapping Tools (GMT) - Unix mapping software, LATEX, Matlab, Ferret, HTML, XML, YAML, KML and Google visualization API.
- Version control systems: Git, SVN.
- Operating Systems: Unix/Linux, Windows and Macintosh.